Assignment - 1

CSL 2050 - Pattern Recognition & Machine Learning

Max. Marks: 50

Submission Instructions:

- Students must submit one Jupyter notebook (.ipynb) file containing:
  - Executable code
  - Visible outputs (tables, figures, metrics)
  - Brief comments or markdown explanations where required
- File naming format : **RollNo.-Name.ipynb** example, 21CS045-AmitSharma.ipynb
- Do Not submit PDFs, word files, or screenshots.
- Late submissions will be penalized as per course policy.
- Date of last submission: **4th February 2026, 11:59PM**.

Part A: Breast Cancer Diagnosis using KNN

Dataset:
https://www.kaggle.com/datasets/gkalpolukcu/knn-algorithm-dataset?select=KNNAlgorithmDataset.csv

Tasks:

1. Perform exploratory analysis i.e., evaluating data distribution, etc. (can use libraries) [5]

2. Implement k-NN algorithm with different ‘k’ values [for ex.: 3, 5, 7] and distance metrics [ex.: Manhattan, Euclidean, etc.]. Compute, visualize and report the accuracies (train, test, validation) for each value of ‘k’ for different distance metrics. (**from scratch without using any libraries**) [5]

3. Perform **K**-fold cross-validation techniques for **K**=5 (such as stratified, random, leave-one-out-once, etc.). Compute and report the mean cross-validation accuracies for different values of **K**. Visualize the average cross-validation accuracies and justify the choice (of **K**) with highest mean cross-validation accuracy. (use libraries) [5]

4. For the best value of **K**, train the k-NN algorithm on the training dataset and evaluate on the test dataset. [5]

5. Why is KNN referred to as a lazy learning algorithm? [5]

Part B: Medical Data Analysis using Regression

Dataset: Diabetes Progression dataset
https://www.kaggle.com/datasets/mathchi/diabetes-data-set?select=diabetes.csv

Tasks:

1. Perform exploratory analysis i.e., names of all features, dataset shape, plotting histogram showing the distribution of the progression score. (use libraries) [5]

2. Implement Simple and Multiple linear regression. (**from scratch by creating a function**) [5]

3. Transform data using PolynomialFeatures (degree = 2) and perform **K**-fold cross-validation techniques for **K**=5 (such as stratified, random, leave-one-out-once, etc.). Compute and report the mean cross-validation accuracies for different values of **K**. Repeat for K=1, 2, 3 and plot MSE vs. polynomial degree. Report optimal degree. (use libraries) [5]

4. Evaluate both models via confusion matrix. Compute and report MAE and R-squared for both the models. Visualize the training progress along the epochs. (use libraries) [5]

5. Discuss the results. [5]